Akoizo Original Practice - Form 4 - Answer key

Division B | Write It Do It | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (4 points) What is the x coordinate of the blue cube center?

Authored Write It Do It case 4. This is a written reconstruction exercise; coordinates describe the target and are not a diagram handed to a builder in competition. A flat 48-cm square base has an origin at its southwest corner. East is +x and north is +y; coordinates are in cm and viewed from above. Put red cube R at (5,9); blue cube B at (11,9); green cube G at (11,21); and yellow cube Y at (5,21). All cubes have edge 1 cm and centers at the named coordinates. A horizontal straw joins the top centers of R and B, a second joins B and G, and a third joins G and Y. No straw joins Y and R. No parts are hidden. The inventory has four cubes, three straws, and the base.

answer: 11 cm. (4 points).

Numerical target 11; absolute tolerance 0.011. Accepted units: , cm.

2. (4 points) What is the y coordinate of the green cube center?

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answer: 21 cm. (4 points).

Numerical target 21; absolute tolerance 0.021. Accepted units: , cm.

3. (4 points) How far east of the red cube center is the blue cube center?

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answer: 6 cm. (4 points).

Numerical target 6; absolute tolerance 0.006. Accepted units: , cm.

4. (4 points) How far north of the blue cube center is the green cube center?

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answer: 12 cm. (4 points).

Numerical target 12; absolute tolerance 0.012. Accepted units: , cm.

5. (4 points) What is the straight-line center-to-center distance from red to yellow?

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answer: 12 cm. (4 points).

Numerical target 12; absolute tolerance 0.012. Accepted units: , cm.

6. (4 points) What is the total center-to-center length of the three specified straw spans?

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answer: 24 cm. (4 points).

Numerical target 24; absolute tolerance 0.024. Accepted units: , cm.

7. (4 points) How many straw connections meet the blue cube?

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answer: 2 . (4 points).

Numerical target 2; absolute tolerance 0.002. Accepted units: , .

8. (4 points) How many straw connections meet the yellow cube?

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answer: 1 . (4 points).

Numerical target 1; absolute tolerance 0.001. Accepted units: , .

9. (4 points) Write unambiguous words-only instructions for establishing the base orientation and placing the red cube.

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answer: Identify the southwest corner as origin, the east edge direction as +x and north as +y. View from above. Place the red cube center 5 cm east and 9 cm north of the origin. State that coordinates locate cube centers, not edges. (4 points).

10. (4 points) Write a complete words-only sequence to place all four cubes and connect the three straws.

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answer: Establish the stated axes; position red at (5,9), blue at (11,9), green at (11,21) and yellow at (5,21) using center coordinates. Connect top center red to blue, blue to green, green to yellow. Leave yellow-to-red unconnected. Numerals used within prose are acceptable for this practice. (4 points).

11. (4 points) A builder places the green cube south of the blue cube. Identify the direction error and give the correction.

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answer: The target green cube is 12 cm north of blue, not south. Keep green's x coordinate equal to blue's and increase its y coordinate by 12 cm. (4 points).

12. (4 points) Explain why “put the other cube beside it” is insufficient for this target.

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answer: It does not identify either cube, the side, orientation, distance or reference point. Use colors/identifiers, a fixed viewpoint and center-to-center offsets. (4 points).

13. (4 points) State two checks that catch an unintended mirror image.

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answer: Check the fixed east/north orientation against the southwest origin; then verify that blue is east of red and green is north of blue. Either independent signed offset check helps detect mirroring. (4 points).

14. (4 points) Describe how you would identify a missing straw without using a drawing in your instructions.

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answer: List the three endpoint pairs explicitly: red-blue, blue-green, green-yellow. Compare that list with the construction and also state that yellow-red must remain open. (4 points).

15. (4 points) Explain the difference between specifying cube centers and specifying the nearest edges.

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answer: They are different reference positions. Each center is half a cube edge from its faces. A dimension must name its reference points consistently or the reconstructed separation shifts. (4 points).

16. (4 points) A builder is missing one cube but connects two later pieces correctly to one another. Explain how to document the useful partial reconstruction.

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answer: Record the correct downstream piece-to-piece relationship separately from the missing or incorrect earlier connection. Do not discard all later relationships solely because an earlier part is wrong; event scoring can recognize correct connections beyond an incorrect one. (4 points).

17. (4 points) Rewrite “left” and “right” in a way that prevents the writer and builder from using opposite viewpoints.

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answer: Specify a common top-down viewpoint and fixed north/east axes on the base, then replace left/right with west/east or signed coordinate offsets relative to a named reference. (4 points).

18. (4 points) Explain how to test the instructions with a partner while avoiding feedback that makes a weak description appear successful.

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answer: Have the partner reconstruct from the written description alone using a fresh kit, without seeing the target or receiving extra verbal hints. Compare the result with the stated locations and connection list, then revise ambiguous steps. (4 points).